

M5 — Joints / constraints

Status: planned.

Goal

Reproduce the three constraint types the game uses, solved in the same constraint loop as contacts.

Scope

- **Revolute** — `PivotJoint` (+ `MotorJoint` for motor, `AngleJoint` for limits), as assembled in `NapeWorld.hx:jointRev`.
- **Weld** — `WeldJoint`, rigid or soft (`stiff` / `frequency` / `damping`).
- **Distance** — `DistanceJoint`, min/max length, soft option.
- `worldPointToLocal` anchor transforms; `space.world` (handle 0) attachment.

Gate

- Each joint type, driven through a representative motion, stays bit-identical for N steps.
- The **level-12 switch/joint** scenario is bit-identical (motor + limits + weld interplay), including `wakeJointPartners` behaviour (see M6).

Risks / open questions

- Soft-constraint frequency/damping → stiffness conversion must match exactly.
- Joint solve ordering relative to contact arbiters within an iteration.